Seminar 9

06.05.2023 10:15

Changing coordinate systems

$\mathcal{R} = (O, \vec{e}_1, \vec{e}_2, \dots, \vec{e}_n) = (O, b)$

$\mathcal{R}' = (O', \vec{e}_1{}', \vec{e}_2{}', \dots, \vec{e}_n{}') = (O', b')$

reference systems for $\mathbb{E}^n$

$P \in \mathbb{E}^n$

$[P]_{\mathcal{R}} = [\vec{OP}]_b$

We want to find $[P]_{\mathcal{R}'}$ with respect to $[P]_{\mathcal{R}}$

$[P]_{\mathcal{R}'} = [\vec{O'P}]_{b'} = [id]_{b,b'} [\vec{O'P}]_b = [id]_{bb'} \cdot [\vec{OP} - \vec{OO'}]_b = [id]_{bb'} [\vec{OP}]_b - [id]_{bb'} \cdot [\vec{OO'}]_b =$

$= [id]_{bb'} \cdot [P]_{\mathcal{R}} - [id]_{bb'} \cdot [O']_{\mathcal{R}} =$

$= [id]_{bb'} [P]_{\mathcal{R}} + [\vec{O'O}]_{b'} =$

$= [id]_{bb'} [P]_{\mathcal{R}} + [O]_{\mathcal{R}'}$

In this course

$[f]_{b,b'} = M_{b',b}(f)$

$[id]_{b,b'} = M_{b',b}(id) = M_{b',b} = M_{\mathcal{R}',\mathcal{R}} =$ "the base change matrix $\mathcal{R}$ to $\mathcal{R}'$"

$[id]_{b,b'} = ([e_1]_{b'}\ [e_2]_{b'} \dots [e_n]_{b'})$

Changing reference frames.

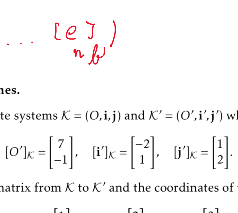
1. We consider two coordinate systems $\mathcal{K} = (O, \mathbf{i}, \mathbf{j})$ and $\mathcal{K}' = (O', \mathbf{i}', \mathbf{j}')$ where

$$[O']_{\mathcal{K}} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{K}} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{K}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Determine the base change matrix from $\mathcal{K}$ to $\mathcal{K}'$ and the coordinates of the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

in the system $\mathcal{K}'$. Further, determine the base change matrix from $\mathcal{K}'$ to $\mathcal{K}$ and use it with the previously obtained coordinates to calculate $[A]_{\mathcal{K}}$, $[B]_{\mathcal{K}}$ and $[C]_{\mathcal{K}}$.



$[O']_K = \begin{pmatrix} 7 \\ -1 \end{pmatrix}, \quad [i']_K = \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \quad [j']_K = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$K = (O, i, j), \quad K' = (O', i', j')$

$M_{KK'} = ([i']_K, [j']_K) = \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix}$

$M_{K'K} = M_{KK'}^{-1} = \frac{1}{-5} \cdot \begin{pmatrix} 2 & -1 \\ -1 & -2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix}$

$[A]_{K'} = M_{K'K} \cdot ([A]_K - [O']_K) = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 7 \\ -1 \end{pmatrix} \right) = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ 3 \end{pmatrix} = \frac{3}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \frac{3}{5} \begin{pmatrix} 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$

$[B]_{K'} = M_{K'K} ([B]_K - [O']_K) = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \left(\begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 7 \\ -1 \end{pmatrix} \right) = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

$[C]_{K'} = M_{K'K} ([C]_K - [O']_K) = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \left(\begin{pmatrix} 3 \\ 1 \end{pmatrix} - \begin{pmatrix} 7 \\ -1 \end{pmatrix} \right) = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 2 \end{pmatrix} = \frac{2}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

!!! $[id]_{bb'} = M_{K'K} = ([e_1]_{b'}, [e_2]_{b'}, \dots [e_n]_{b'})$

$[P]_{K'} = M_{K'K} ([P]_K - [O']_K) = M_{KK'} \cdot [P]_K + [O]_{K'}$

2. With the assumptions in the previous exercise, give parametric equations and Cartesian equations for the lines AB, AC, BC both in the coordinate system $\mathcal{K}$ and in the coordinate system $\mathcal{K}'$.

[1] $\vec{AB}_K = [B]_K - [A]_K = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \Rightarrow AB_K : \begin{cases} x = x_A + at = 1 + t \\ y = y_A + bt = 2 + 2t \end{cases}$

$AB_{K'} : \begin{pmatrix} x' \\ y' \end{pmatrix} = M_{K'K} \cdot \left[\begin{pmatrix} x \\ y \end{pmatrix} - [O']_K \right] = \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 + t - 7 \\ 2 + 2t + 1 \end{pmatrix} = \frac{1}{5} \cdot \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ 3 \end{pmatrix} + \frac{1}{5} \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} t = \begin{pmatrix} 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} t$

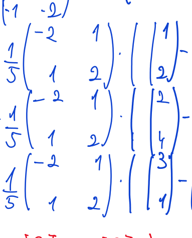
The same formulas as the one above apply here

[2] $\vec{AC}_K = [C]_K - [A]_K = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \Rightarrow AC_K : \begin{cases} x = x_A + 2t \\ y = y_A - t \end{cases} \Leftrightarrow \begin{cases} x = 1 + 2t \\ y = 2 - t \end{cases}$

$\vec{AC}_{K'} = [C]_{K'} - [A]_{K'} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \Rightarrow AC_{K'} : \begin{cases} x' = x'_A - t \\ y' = y'_A \end{cases} \Leftrightarrow \begin{cases} x' = 3 - t \\ y' = 0 \end{cases}$

[3] $\vec{BC}_K = \vec{BA}_K + \vec{AC}_K = \begin{pmatrix} -1 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} : \begin{cases} x = 2 + t \\ y = 4 + t \end{cases}$

$\vec{BC}_{K'} = \vec{BA}_{K'} + \vec{AC}_{K'} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} : \begin{cases} x' = 3 - t \\ y' = 1 - t \end{cases}$



3. Consider the tetrahedron $ABCD$ and the coordinate systems

$$\mathcal{K}_A = (A, \vec{AB}, \vec{AC}, \vec{AD}), \quad \mathcal{K}'_A = (A, \vec{AB}, \vec{AD}, \vec{AC}), \quad \mathcal{K}_B = (B, \vec{BA}, \vec{BC}, \vec{BD}).$$

Determine

a) the coordinates of the vertices of the tetrahedron in the three coordinate systems,

b) the base change matrix from $\mathcal{K}_A$ to $\mathcal{K}'_A$,

c) the base change matrix from $\mathcal{K}_B$ to $\mathcal{K}_A$.

a) $[A]_{K_A} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [B]_{K_A} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad [C]_{K_A} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad [D]_{K_A} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$[A]_{K_{A'}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [B]_{K_{A'}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad [C]_{K_{A'}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad [D]_{K_{A'}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$[A]_{K_B} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad [B]_{K_B} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [C]_{K_B} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad [D]_{K_B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

b) $M_{K'_A K_A} = [\,[e_1]_{K_{A'}}\ [e_2]_{K_{A'}}\ [e_3]_{K_{A'}}\,] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

c) $M_{K_A K_B} = [\,[e_1]_{K_B}\ [e_2]_{K_B}\ [e_3]_{K_B}\,] = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

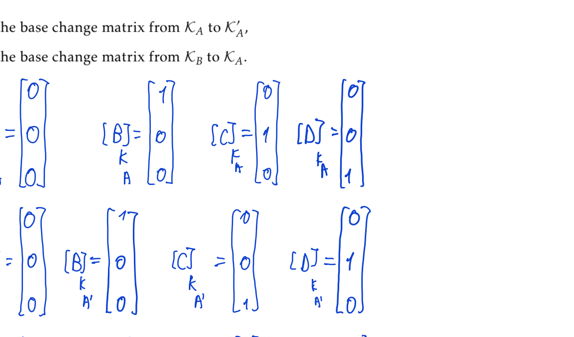
4. We consider the coordinate systems $\mathcal{K} = (O, \mathbf{i}, \mathbf{j}, \mathbf{k})$ and $\mathcal{K}' = (O', \mathbf{i}', \mathbf{j}', \mathbf{k}')$ where

$$[O']_{\mathcal{K}} = \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{K}} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{K}} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad [\mathbf{k}']_{\mathcal{K}} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

Determine the base change matrix from $\mathcal{K}$ to $\mathcal{K}'$ and the coordinates of the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 5 \\ 1 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} -3 \\ 7 \\ 1 \end{bmatrix}, \quad [D]_{\mathcal{K}} = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}.$$

in the coordinate system $\mathcal{K}'$. Further, determine the base change matrix from $\mathcal{K}'$ to $\mathcal{K}$ and use it with the previously determined coordinates to calculate $[A]_{\mathcal{K}}$, $[B]_{\mathcal{K}}$, $[C]_{\mathcal{K}}$ and $[D]_{\mathcal{K}}$.



$M_{KK'} = [\,[i']_K\ [j']_K\ [k']_K\,] = \begin{pmatrix} -1 & -2 & 0 \\ -2 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

$\det(M_{KK'}) = \begin{vmatrix} -1 & -2 & 0 \\ -2 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = -2 - 8 = -10$

$M_{KK'}^T = \begin{pmatrix} -1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$

$M_{K'K} = M_{KK'}^{-1} = \frac{1}{-10} \begin{pmatrix} 2 & 4 & -2 \\ 4 & -2 & 1 \\ 0 & 0 & -5 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} -2 & -4 & 2 \\ -4 & 2 & -1 \\ 0 & 0 & 5 \end{pmatrix}$

$[A]_{K'} = M_{K'K} ([A]_K - [O']_K) = \frac{1}{10} \begin{pmatrix} -2 & -4 & 2 \\ -4 & 2 & -1 \\ 0 & 0 & 5 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -1 \\ 0 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 10 \\ 10 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$[B]_{K'} = M_{K'K} ([B]_K - [O']_K) = \frac{1}{10} \cdot \begin{pmatrix} -2 & -4 & 2 \\ -4 & 2 & -1 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} -3 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 10 \\ 10 \\ 10 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$[C]_{K'} = M_{K'K} ([C]_K - [O']_K) = \frac{1}{10} \begin{pmatrix} -2 & -4 & 2 \\ -4 & 2 & -1 \\ 0 & 0 & 5 \end{pmatrix} \cdot \begin{pmatrix} -7 \\ 2 \\ 2 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 10 \\ 30 \\ 10 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$

$[D]_{K'} = M_{K'K} ([D]_K - [O']_K) = \frac{1}{10} \begin{pmatrix} -2 & -4 & 2 \\ -4 & 2 & -1 \\ 0 & 0 & 5 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -2 \\ 2 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 20 \\ 10 \\ 10 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

5. With the assumptions in the previous exercise, give parametric equations and Cartesian equations for the line AB and the plane ACD both in the coordinate system $\mathcal{K}$ and in the coordinate system $\mathcal{K}'$.

$\vec{AB}_K = [B]_K - [A]_K = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} : \begin{cases} x = 1 \\ y = 4 + t \\ z = -1 + 2t \end{cases}$

$\vec{AB}_{K'} = [B]_{K'} - [A]_{K'} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} : \begin{cases} x = 1 \\ y = 1 \\ z = t \end{cases}$

$\vec{AC}_K = [C]_K - [A]_K = \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix}$

$\vec{AD}_K = [D]_K - [A]_K = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$

$\begin{vmatrix} x & y - 3 & z - 1 \\ -4 & 3 & 2 \\ -1 & -1 & 2 \end{vmatrix} = 6x + 4z - 4 - 2y + 6 + 3z - 3 + 2x + 8y - 12 = 8x + 6y + 7z - 13 = 0$

$\begin{cases} x = -4\lambda - \mu \\ y = -3 + 3\lambda - \mu \\ z = -1 + 2\lambda + 2\mu \end{cases}$

6. Consider $\mathbf{v}(2,1,1) \in \mathbb{V}^3$ and $Q(2,2,2) \in \mathbb{E}^3$.

a) Give the matrix form for the parallel projection on the plane $\pi : z = 0$ along the line $Q + \langle \mathbf{v} \rangle$.

b) Give the matrix form for the parallel reflection in the plane $\pi : z = 0$ along the line $Q + \langle \mathbf{v} \rangle$.

7. Determine the orthogonal projection of the point $A(2, 11, -5)$ on the plane $x + 4y - 3z + 7 = 0$ by determining the matrix form of the projection. (Compare your result with the previous seminar.)

6) a) $e(x,y,z) = z$

$\text{lin}\, e(x,y,z) = z$

$\vec{a} = (0,0,1)$

$\vec{v} = (2,1,1)$

$\text{lin}\, e(x_1, x_2, \dots, x_n) = a_1 x_1 + \dots + a_n x_n$

$[Pr_{H,\vec{v}}(P)]_K = \frac{1}{\text{lin}\, e(v)} \cdot ((v^t \cdot a) \cdot I_n - v \otimes a) [P]_K - \frac{a_{n+1}}{\text{lin}\, e(v)} [\vec{v}]_K$

$= \left(I_n - \frac{v \cdot a^t}{v^t \cdot a} \right) \cdot [P]_K - \frac{a_{n+1}}{v^t \cdot a} [\vec{v}]_K$

$[Ref_{H,\vec{v}}(P)]_K = 2 \cdot [Pr_{H,\vec{v}}(P)]_K - [P]_K$

$[Pr_{H,\vec{v}}(Q)]_K = \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \right) \cdot \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}$

$[Ref_{H,\vec{v}}(P)]_K = 2 \cdot \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -6 \\ -2 \\ -2 \end{pmatrix}$

7) $A(2, 11, -5)$

$x + 4y - 3z + 7 = 0 \quad \nearrow d(A, \pi) = \frac{|2 + 44 + 15 + 7|}{\sqrt{150}}$

$H: x + 4y - 3z + 7 = 0$

$\vec{v}(1, 4, -3)$

$A(2, 11, -5)$

$\vec{a}(1, 4, -3)$

$\begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix} \cdot (1\ \ 4\ \ -3) = \begin{pmatrix} 1 & 4 & -3 \\ 4 & 16 & -12 \\ -3 & -12 & 9 \end{pmatrix}$

$(1\ \ 4\ \ -3) \cdot \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix} = 26$

$[Pr_{H,\vec{v}}(A)]_K = \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{26} \begin{pmatrix} 1 & 4 & -3 \\ 4 & 16 & -12 \\ -3 & -12 & 9 \end{pmatrix} \right) \cdot \begin{pmatrix} 2 \\ 11 \\ -5 \end{pmatrix} - \frac{7}{26} \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$

$= \frac{1}{26} \left(\left(\begin{pmatrix} 26 & 0 & 0 \\ 0 & 26 & 0 \\ 0 & 0 & 26 \end{pmatrix} - \begin{pmatrix} 1 & 4 & -3 \\ 4 & 16 & -12 \\ -3 & -12 & 9 \end{pmatrix} \right) \begin{pmatrix} 2 \\ 11 \\ -5 \end{pmatrix} - 7 \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix} \right) =$

$= \frac{1}{26} \begin{pmatrix} -16 \\ -28 \\ 74 \end{pmatrix}$

8. Determine the orthogonal reflection of the point $P(6, -5, 5)$ in the plane $2x - 3y + z - 4 = 0$ by determining the matrix form of the reflection. (Compare your result with the previous seminar.)

$\pi: 2x - 3y + z - 4 = 0$

$\vec{v}(2, -3, 1)$

$\vec{a}(2, -3, 1), \ a_4 = -4$

$[Pr_{H,\vec{v}}(P)]_K = \left(I_3 - \frac{v \cdot a^t}{v^t \cdot a} \right) \cdot [P]_K - \frac{a_{n+1}}{v^t \cdot a} \cdot [\vec{v}]_K$

$v \cdot a^t = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \cdot (2\ \ -3\ \ 1) = \begin{pmatrix} 4 & -6 & 2 \\ -6 & 9 & -3 \\ 2 & -3 & 1 \end{pmatrix}$

$v^t \cdot a = (2\ \ -3\ \ 1) \cdot \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = 14$

$[Pr_{H,\vec{v}}(P)]_K = \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{14} \cdot \begin{pmatrix} 4 & -6 & 2 \\ -6 & 9 & -3 \\ 2 & -3 & 1 \end{pmatrix} \right) \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \frac{4}{14} \cdot \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$

9. Determine the orthogonal projection of the line $\ell : 2x - y - 1 = 0 \cap x + y - z + 1 = 0$ on the plane $\pi : x + 2y - z = 0$ by determining the matrix form of the projection. (Compare your result with the previous seminar.)

$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \quad \vec{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$\vec{v} \cdot \vec{a}^t = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \cdot (1\ \ 2\ \ -1) = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}$

$\vec{v}^t \cdot \vec{a} = (1\ \ 2\ \ -1) \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 6$

$e(x,y,z) = x + 2y - z$

$\text{lin}\, e(x,y,z) = x + 2y - z$

$[Pr^{\perp}_{\pi}(P)] = \frac{1}{\text{lin}\, e} (\vec{v}^t \cdot \vec{a}\, I_3 - \vec{v} \cdot \vec{a}^t) [P] - \frac{a_{n+1}}{\text{lin}\, e} v$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 5 & -2 & 1 \\ -2 & 2 & 2 \\ 1 & 2 & 5 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\ell : \begin{cases} 2x - y = 0 \\ x + y - z = 0 \end{cases} \Leftrightarrow \begin{cases} x = t \\ y = 2t - 1 \\ z = 3t \end{cases}$

$Pr^{\perp}_{\pi}(\ell) : \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 5 & -2 & 1 \\ -2 & 2 & 2 \\ 1 & 2 & 5 \end{pmatrix} \cdot \begin{pmatrix} t \\ 2t - 1 \\ 3t \end{pmatrix} \Leftrightarrow$

$\Leftrightarrow \begin{cases} x = \frac{1}{3}(2t + 1) \\ y = \frac{1}{3}(4t - 1) \\ z = \frac{1}{3}(10t - 1) \end{cases}$

10. Give Cartesian equations for the line passing through the point $M(1, 0, 7)$, parallel to the plane $\pi : 3x - y + 2z - 15 = 0$ and intersecting the line

$$\ell : \frac{x-1}{4} = \frac{y-3}{2} = \frac{z}{1}.$$

$\vec{v}(a, b, c)$

$\pi : 3x - y + 2z - 15 = 0 \Rightarrow \vec{n}(3, -1, 2)$

$\vec{v} \parallel \pi$

$\vec{v} \nparallel \ell$

$\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ v_x & v_y & v_z \\ u_x & u_y & u_z \end{vmatrix} = 0 \Leftrightarrow \vec{u} \nparallel \vec{v}$

$\ell : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$

$\ell_1 : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$

$\Rightarrow \begin{vmatrix} 0 & 3 & -7 \\ 4 & 2 & 1 \\ v_x & v_y & v_z \end{vmatrix} = 0$

$-28 v_y + 3 v_x + 14 v_x - 12 v_z = 0$

$\begin{cases} 17 v_x - 28 v_y - 12 v_z = 0 \\ 3 v_x - v_y + 2 v_z = 0 \quad | \cdot 6 \end{cases}$

$v_x = \alpha$

$\begin{cases} 17\alpha - 28 v_y - 12 v_z = 0 \\ 18\alpha - 6 v_y + 12 v_z = 0 \end{cases} \Leftrightarrow \begin{cases} v_y = \frac{35}{34}\alpha \\ v_z = \frac{3\alpha - \frac{35}{34}\alpha}{2} = \frac{67}{68}\alpha \end{cases}$

$\frac{x - 1}{1} = \frac{y}{\frac{35}{34}} = \frac{z - 7}{\frac{67}{68}}$